Consider an infinite surface filling the y-z plane at $x = -1$ (m), with uniform surface charge density ρ_s=1 (mC/m^2). Calculate the potential difference between points A(2, 0, 4) and B(7, -3, 1).

The potential function due to some charge arrangement is given by

$$V(\rho, \varphi, z) = 3\,\rho\, e^{-z} \cos\varphi \qquad \text{(V)}$$

a) Calculate the potential difference between points A(5, 30°, 0) and B(0, 45°, -2).
b) Find the electric field intensity at point B.
c) Find the total energy stored in the volume specified by $0 \le \rho, z \le 1$.

$$\int (\cos^2 x)\, dx = \left(\frac{x}{2} + \frac{\sin 2x}{4}\right) + C$$

Consider a charged, zero-thickness, circular ring of radius a with <u>non-uniform</u> line charge density given by

$$\rho_l = \rho_o \sin\varphi \ \text{(C/m)} \qquad (\rho_o \text{ is a constant})$$

Use Coulomb's law to derive an expression for the electric field intensity produced by the ring at any point along its axis.

The electrostatic flux density within a solid sphere of radius 5 m is given by:

$$\vec{D} = \frac{20}{r}\,\vec{a}_\varphi + \frac{100}{r^2}\,\vec{a}_r \quad \text{(C/m}^2\text{)}$$

The center of the sphere coincides with the origin.

a) Find the total charge enclosed in the sphere.
b) Calculate the volume charge density within the sphere.

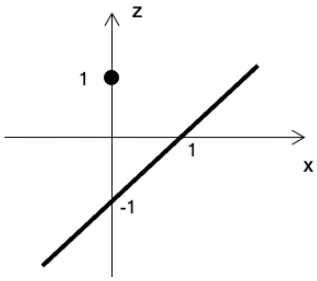
The electric field intensity in a certain space is given by

$$\vec{E} = \frac{r}{10}\,\vec{a}_r \qquad V/m$$

Given the electric potential at the origin as 2 V,

a) Find the electric potential at point *P*(3,45°,0°). Show all steps.
b) At what point(s), other than the origin will the potential equal to zero? Explain.

Consider two infinite charge-carrying structures placed as in the figure below. Structure 1 is an infinite line carrying uniform charge density ρ_L=100 nC/m and structure 2 is an infinite sheet carrying uniform charge density ρ_s=-5 nC/m^2.



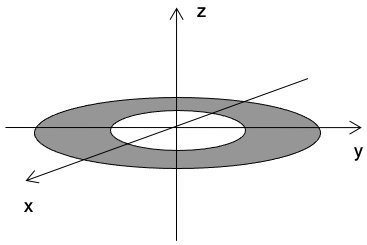
1) Find the total electric field intensity at the origin.
2) Find the total electric field intensity at the point P(2, 1, 0).

Using Gauss's law, find general expressions for the electric potential everywhere in space due to a charged sphere. The sphere has a radius R and carries a non-uniform volume charge density ρ_v. The charge density is given by

$\rho_v = 2\beta e^{-\alpha r}$ for $0 \leq r \leq R$, α and β are positive constants.
$\rho_v = 0$ elsewhere.

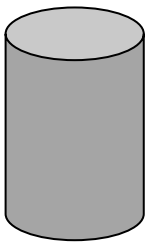
Sketch the electric field and the electric potential versus radial distance.

Use Coulomb's law to derive an expression for the electric field intensity along the z-axis due to the charged structure shown below. The structure has inner radius a and outer radius b and carries a uniform charge density ρ_s (C/m^2).



1) What is value of the electric field intensity at the origin?
2) Sketch the magnitude of the field intensity versus z.

Consider a thick infinite rod charged with a uniform charge density ρ_v. The rod has a circular cross section of radius a with axis coinciding with the z-axis.



1) Use Gauss's Law to find the electric field intensity at any point inside or outside the rod.
2) Plot the magnitude of the electric field as a function of distance away from the z-axis.

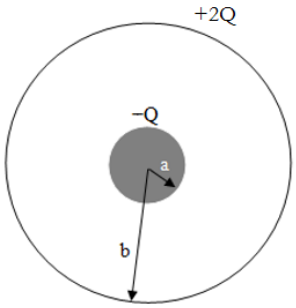
The potential function inside a piece of glass due to some charge arrangement is given by

$$V(x, y, z) = 2ze^{-2x} - 3y \quad \text{(V)}$$

1) Calculate the potential difference between point A(5, 1, 0) and point B(0, 1, -2).
2) Find the electric field intensity at point B.
3) Calculate the total charge enclosed in the volume specified by $0 \leq x \leq 1$, $0 \leq y \leq 1$ and $-1 \leq z \leq 1$.

4) Find the total energy stored in the volume of part (3).

Consider a conducting solid sphere of radius a, carrying a charge -Q and surrounded by a very thin conducting spherical shell of radius b with charge +2Q (see figure).



a) Find the electric field intensity everywhere using Gauss's law.
b) Sketch the magnitude of the field versus distance along the radial direction. Label all important points
c) Find the location of maximum field magnitude.

Consider a charged, zero-thickness, circular ring of radius R with **uniform** line charge density ρ_L. The ring is centered at the origin and is placed in the $y-z$ plane.

a) Use Coulomb's law to derive an expression for the electric field intensity produced by the ring at any point along the *x*-axis.
b) For R=10 cm and ρ_L=1 nC/m, calculate the field intensity at the cylindrical coordinate point (10 cm, -180°, 0).
